**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 228 · Object-oriented synoptic challenge

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the difference between Step Into and Step Over?

Q2. What is encapsulation?

Q3. How would you store many Book objects in C#?

Q4. A Borrow method should fail on a book already on loan. Give a useful test.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

class — Template defining properties and methods.

object — Instance of a class.

encapsulation — Control access to and modification of object state.

collection — Stores multiple elements, such as List<Book>.

invariant — A property that should stay true while an object is used.

## New learning

Model the entity — A Book has a Title and whether it is currently on loan.

Keep state protected — The public Borrow method updates a private-set OnLoan property, so external code cannot change it freely.

Test a second call — The first Borrow returns true and changes state; the second call returns false because the book is already on loan.

## Worked example

Model a Seat object. Reserve() should succeed once, then fail.

1. Predict — Initial Reserved is false, so Reserve sets it to true and returns true.

2. Model response — The second call detects Reserved already true; it returns false and leaves state unchanged.

# We Do · Practise with guidance

A Locker class starts Occupied=false. What happens after Use(), then Use() again?

```
bool Use() {
 if(Occupied) return false;
 Occupied=true; return true;
}
```

Predict

The first call sets Occupied to true and returns true.

Model response

The second call returns false without changing Occupied.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain the difference between a class and an object using Book.

### Question 2

Write a method ReturnBook() that only succeeds if OnLoan is true, then explain one test.

### Question 3

Explain how to manage many Book objects, retrieve a book by unique ISBN, and test duplicate borrowing.

## Exit Ticket · Answer before checking

Two Book objects created from the same class must always share the same OnLoan value.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Step Into enters the called method; Step Over executes it without stopping inside.

Q2. Controlling how an object's internal state can be accessed or changed.

Q3. For example List<Book>, a collection holding Book references.

Q4. Set OnLoan=true, call Borrow, expect false and OnLoan to stay true.

### You Do pages · Independent questions

Q1. [4] A class is a blueprint/type [1] that defines attributes/methods [1]; an object is a particular instance [1], e.g. a Book for one title [1].

Q2. [6] public bool ReturnBook(){ if(!OnLoan)return false; OnLoan=false; return true; } [4]; test after a successful Borrow, ReturnBook returns true [1] and OnLoan becomes false [1].

Q3. [6] Store Book objects in a Dictionary<string,Book> keyed by ISBN [2]; unique ISBN maps to one Book [1]; look up the selected object [1]; borrow once succeeds [1], borrowing again returns false and state remains on loan [1].

### Exit Ticket · Model answer

False. Each object maintains its own state.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**



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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check my understanding